Atlas 210X — VFO Nano add-on wiring

YT3CT (ex YU5KBM) · Nano ATmega328P (Old Bootloader) · sketch atlas_vfo_fixed.ino · 2026-10 · button / relay part = proposal

1. LCD RG1602A (16×2) — 5 twisted pairs

Pair	Wire A (Nano → 470 Ω → LCD pin)	Wire B (twisted with A)
1	D8 → E → 6	GND → 1
2	D7 → RS → 4	GND → 1
3	D9 → D4 → 11	D10 → D5 → 12
4	D11 → D6 → 13	D12 → D7 → 14
5	+5 V → VDD 2 (no R)	GND → 1

- **At the LCD (not in the cable):** R/W 5 → 1; backlight 15 → 2, 16 → 1 (100 Ω on board); 10k contrast pot 1–3–2 (wiper to V0 pin 3); 100 nF + 10 µF across 2–1. D0–D3 (7–10) open.
- 470 Ω series R at the **Nano end** (220 Ω–1 k OK). All 3 GND wires to the Nano GND pin next to D2; LCD not grounded to the chassis.
- ~1 twist / 1–2 cm, slightly different rate per pair, < 1 cm untwisted at the ends. Route along the chassis, cross other wiring at 90°, clip-on ferrite near the Nano.

2. TTL programming leads (no USB needed)

USB-TTL cable (Prolific)	Nano
Green (TX)	D0 / RX
White (RX)	D1 / TX
Black (GND)	GND
Red (+5 V)	**NOT connected**
—	RST lead (+ optional push button to GND)

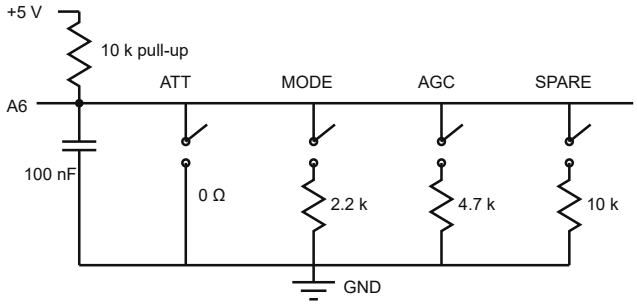
- If no sync: swap green / white (colours differ between cables).
- **Upload:** start upload, then short RST to GND for ½ s. Power cycling only works if the radio stays off until the Nano POW LED is fully out (TTL line back-feeds the Nano).
- Same procedure for the VariBeam Nano.
- Never plug the Nano USB in while the TTL cable is connected.

Free Nano pins & plan

Pin	Now	Plan
D0 / D1	Serial	TTL leads
D4	free	spare
D5	free	ATT relay driver
D13	free (LED)	AGC fast (2N7000)
A6	free (analog only)	button ladder
A7	free (analog only)	spare / 2nd ladder

Used: D2/D3 encoder, D6 enc. button, D7–D12 LCD, A0/A1 S-meter, A2 TX, A3 band, A4/A5 I²C Si5351.

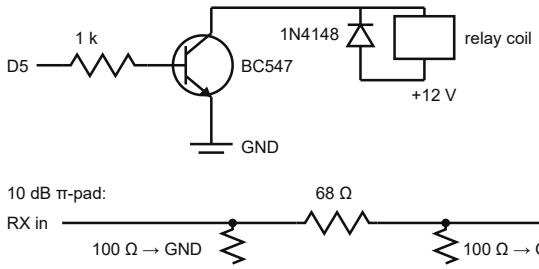
3. Buttons on A6 (resistor ladder)



Button	R to GND	ADC	Accept if
ATT on/off	0 Ω (direct)	0	< 90
MODE LSB/USB	2.2 k	184	90–255
AGC fast/slow	4.7 k	327	255–420
SPARE	10 k	512	420–770
none	—	1023	> 770

A6/A7 have **no internal pull-up** → the external 10 k is required. One button at a time. Resistors 1 % or 5 % both fine.

4. Attenuator relay (D5) — RX path only



- π-pad 100 / 68 / 100 Ω ≈ 10 dB (exact: 96 / 71 / 96 Ω). Relay bypasses or inserts it.
- **Only in the receive line after the antenna relay — never in the TX path.** Short leads / thin coax.
- 5 V relay: coil to +5 V instead of +12 V. Diode cathode (band) to + side.

5. AGC fast/slow (D13) — later

- 2N7000: gate via 100 Ω from D13, 100 k gate–GND, source GND, drain via R_{fast} (~150–330 k, to try) to the AGC hold node on PC-300D.
- Parts on PC-300D to be identified by value first (hold cap / 1 M discharge R).

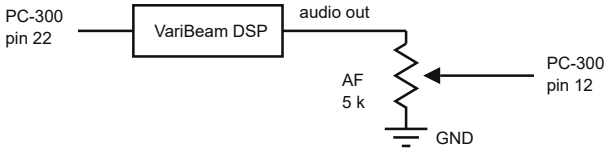
6. MODE LSB/USB

Software only: BFO moved to the other side of the crystal filter (5.520.000 ↔ ~5.523.3 MHz, to be set) — LO calculation follows automatically.

7. Front panel gain controls — 2 concentric dual pots (2 shafts each) values to confirm

Pot	Gang	Function	Value
1	a	AF gain	5 k
	b	RF gain	10 k
2	a	MIC gain	10 k
	b	ALC	20 k

Values as fitted; which shaft (inner/outer) drives which gang, and the original Atlas values, to be measured when the new front plate goes in. RF / MIC / ALC: existing wires moved lug for lug (CW end, wiper, CCW end).



- AGC / S-meter pickoff is before pin 22 → DSP and AF gain do not change the S-meter.
- Shielded cable for the audio leads, shield grounded at one end.